arXiv:2606.10051v2 [cs.CY] 22 Oct 2026

Empirically Grounded Behavioral Adaptivity for LLM-Based Virtual Patients in Psychotherapy Training

Angela H. Chen[1] **Siwei Jin**[2,*] **Catherine Bao**[3,*] **Canwen Wang**[1]
Yi-Chia Wang[1] **Robert Kraut**[1] **Tongshuang Wu**[1] **Haiyi Zhu**[1,†]
[1]Carnegie Mellon University [2]Independent Researcher [3]University of Utah
{angelachen, haiyiz}@cmu.edu

Abstract

Simulated patients offer a scalable way to train psychotherapy micro-skills such as empathic responding and exploratory probing, but current systems either follow fixed scripts or rely on LLMs that drift unpredictably over long sessions. We present the Adaptive Virtual Patient (AVP), which adapts its disclosure behavior in response to trainee skill, from guarded, through moderate openness, to full disclosure. The AVP is informed by a structural equation model fit to nearly 2,000 hours of real-world psychotherapy transcripts, which quantifies how therapist empathy and exploration shift a patient's openness over time. An LLM generates the AVP's utterances conditioned on a disclosure level that the dynamics module updates each turn. In an evaluation with 20 clinicians and trainees over 80 sessions (1,033 turns), the AVP's disclosure rises in response to therapist empathy and exploration, while a prompt-only baseline stays flat; ablations confirm that the empirically motivated parameterization outperforms alternatives, with exploration carrying most of the adaptive signal.

1 Introduction

Large language models (LLMs) are increasingly used as conversational agents in education, training, and social-interaction research (Shao et al., 2023; Martynova et al., 2025; Park et al., 2023). They produce fluent and plausible responses, but their long-horizon behavior is governed by a static persona prompt, which limits the within-session change (Wang et al., 2024a; Samuel et al., 2025). An agent may sound like a patient, student, or customer on each turn but fail to adapt in ways that reflect an evolving state in relation to what the user does (Abdulhai et al., 2026; Li et al., 2024). We study this problem in psychotherapy training,

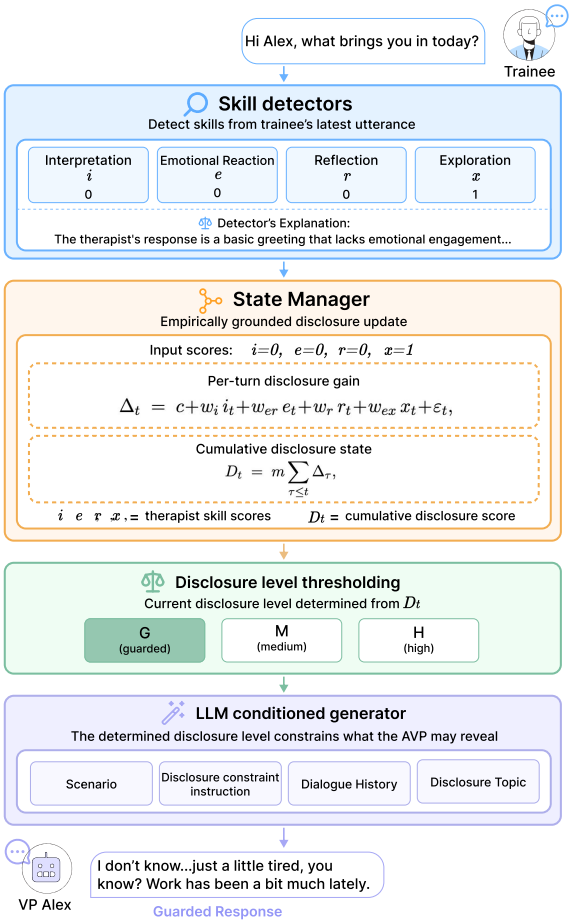


Figure 1: Overview of our Adaptive Virtual Patient (AVP) framework, where we explicitly control the VP's behavioral state via an explicit intermediate computation, such that the VP's openness directly ties with the trainee performance.

where agents simulate patients with different mental health conditions (Kenny et al., 2007; Tanana et al., 2019).

A realistic session with a guarded patient unfolds like this: if the trainee asks closed, agenda-driven questions, the patient gives brief, surface-level answers; if the trainee follows up with open-ended probes ("what was that like for you?") and empathic reflections ("that must have been hard"), the

*Equal contribution; work done at CMU.
†Corresponding author.

patient begins to share more personal, emotionally loaded details. This contingent response is itself feedback for the trainee; without it, the trainee cannot learn what works (Hattie and Timperley, 2007; Mcgaghie et al., 2011).

Despite the promise of using simulated virtual patients (VPs) to scale up the psychotherapy training, existing simulation approaches only partially support this adaptive cycle. Scripted VPs are reproducible but their predefined branches limit turn-by-turn adaptation (Woodham et al., 2015; Kononowicz et al., 2019a); While prompt-only LLM patients support open, fluent exchanges (Lozoya et al., 2025; Wang et al., 2024a), their behavioral change emerges implicitly from the prompt rather than from the trainee's input. Recent adaptive-VP work modifies patient behavior using trainee evaluation scores (Lee et al., 2025), but its rule-based mapping from scores to behavior is not grounded in psychotherapy theory or data and thus does not model the gradual progression by which therapist behaviors shape a patient's state over time.

In response, we propose an **Adaptive VP** framework that separates *behavioral state transitions* from *language generation*. As shown in Figure 1, after each trainee turn, LLM evaluators score clinically meaningful behaviors (empathy and exploration); a dynamics layer accumulates these scores into an ordinal disclosure level (low, medium, high); and an LLM generates the VP's next utterance conditioned on that level. Making the behavioral state an explicit intermediate computation keeps the VP's adaptation controllable and testable.

Specifically, we design the core dynamics layer to be informed by a structural equation model (SEM) estimated from over 2,000 hours of human-to-human psychotherapy session transcripts, which gives directed effects of therapist behaviors on next-turn patient disclosure (Chen et al., 2026). The deployed system uses integer weights approximating the SEM-implied empathy-to-exploration ratio. Self-disclosure is the controlled behavioral construct: the deployed system's discrete disclosure level can be checked against an independent rater's coding turn by turn.

We evaluate the framework in a within-subjects experiment: 20 clinicians and trainees held 80 brief conversations (1,033 turns) with four virtual patients varying in persona and system type. We test whether trainee behaviors produce coherent disclosure trajectories, whether generated utterances align with the intended disclosure level, and whether the patient is perceived as realistic, adaptive, and useful for training. With the adaptive virtual patient, disclosure climbs coherently across the session and matches the system's internal state in the generated text — neither of which the prompt-only baseline achieves.

Contributions. We make three contributions. First, we introduce a virtual-patient framework that handles **behavioral change** as a small, explicit module between the trainee's turn and the VP's reply: it updates a latent disclosure state using weights derived from a psychotherapy-data structural equation model, and the LLM generates the next utterance conditioned on that state. Keeping the behavioral logic separate from generation makes the VP's adaptation inspectable, controllable, and fixed across experimental conditions in a way a prompt-only system cannot achieve.

Second, we propose **four measurable properties** an AVP should satisfy: *P1 Responsiveness* (the state responds to user behavior in the direction the data predicts), *P2 Gradual Build-up* (the state climbs steadily across the session and only as the user earns it), *P3 Faithful Realization* (the generated text matches the system's intended state), and *P4 Skill Sensitivity* (whether disclosure varies with user skill); each property comes with its own metric, so we can locate exactly where a system fails rather than collapse the question into one number.

Third, we provide **empirical evidence** from 20 clinicians and trainees, that the adaptive patient can produce a steadily climbing disclosure trajectory while a prompt-only baseline with the same LLM and persona stays flat, which better support the loop needed for psychotherapy training. On post-session ratings, the adaptive version of an expressive persona outscored its static counterpart on character consistency, adaptivity, realism, and training usefulness, with the strongest endorsements from licensed therapists; and a parameter ablation further shows the deployed weighting beats every alternative tested in agreement with independent ratings.

2 Background and Related Work

Virtual patients and LLM-based role-play agents. Virtual and simulated patients have long been used as a mechanism help trainees practice clinical communication skills such as empathic responding, reflective listening, and exploratory questioning (Barrows, 1993; Kenny et al., 2007; Tanana et al., 2019). Reviews and

meta-analyses show that virtual patients can support health-professions training at scale (Cook and Triola, 2009; Cook et al., 2011; Consorti et al., 2012; Kononowicz et al., 2019b). Traditional systems often prioritize structured case progression and reproducibility, while recent LLM-based systems enable more fluent and open-ended patient simulation through prompt-based personas, theory–informed cognitive models, or domain-expert authored role-plays (Wang et al., 2024b; Holderried et al., 2024; Louie et al., 2024; Lozoya et al., 2025). More general generative-agent architectures further show that memory, reflection, and planning can support believable long-horizon behavior (Park et al., 2023). Across these approaches, however, behavioral change is typically governed by scripts, prompts, memory, or qualitative principles rather than empirically estimated interaction dynamics. Our work instead updates a latent disclosure state from detected trainee behaviors and conditions language generation accordingly.

Controllable dialogue generation. Prior work in controllable dialogue generation has shown that external conditioning signals (e.g., personas, control codes, styles, attributes, and latent discourse variables) can steer generated responses (Zhang et al., 2018; Keskar et al., 2019; Zhao et al., 2017; Serban et al., 2017). These methods generate language conditioned on a desired persona, style, or latent variable, but generally do not specify how that control variable should update as a function of an interlocutor's behavior. Recent role-playing benchmarks similarly evaluate whether LLMs can maintain assigned roles or personas over multi-turn interactions (Wang et al., 2024a; Samuel et al., 2025). In contrast, our setting requires behavior-contingent control: the target state (i.e., the VP's willingness to self-disclose) itself must evolve over time, and those changes should be inspectable and grounded in an empirical model of interaction dynamics. Our framework provides this control through an explicit state-transition step: detected user behaviors update a latent disclosure state before the next VP utterance is generated.

State-based dialogue management. State-based dialogue systems provide a precedent for separating state estimation from response or action generation. In task-oriented dialogue, dialogue state tracking (DST) estimates the values of conversation-relevant slots (e.g., a restaurant's cuisine, price range, or location), user goals, or belief states from conversation history, and downstream policies use those states to select system actions (Young et al., 2013; Balaraman et al., 2021; Jacqmin et al., 2022). This separation makes dialogue behavior more controllable and diagnosable than direct response generation alone. Our work adapts this state-based view to simulated interpersonal behavior; the state is not a task slot or user goal, but a continuous latent behavioral variable, VP disclosure level, whose transition is driven by detected user behaviors and parameterized by empirical psychotherapy data. The next VP utterance is then generated conditioned on this updated state.

3 Method

3.1 Empirically Grounded Adaptivity

We use "empirically grounded adaptivity" to mean four things across a session (elaborated and quantified in §4.1): (i) the VP's internal state changes from turn to turn in response to user behaviors, in a direction consistent with an external data-driven model of interaction dynamics; (ii) the state trajectory climbs gradually, advancing only as user competence accumulates; (iii) generated utterances reflect the internal state rather than running independently of it; and (iv) different user strategies produce different trajectories.

Constructs and Their Measurement To operationalize the adaptivity, our framework adds one explicit step between user input and VP output: an estimate of how the VP's willingness to self-disclose should change.

More specifically, in our system the **user behaviors** are empathy and exploration scores:

- *Empathy*: User behaviors communicating understanding of the VP, scored on a 0-2 scale along three components: interpretation (i) and emotional reaction (e) from EPITOME (Sharma et al., 2020), and reflection (r) from MI (Miller and Rollnick, 2023).
- *Exploration*: User behaviors that invite the VP to elaborate, such as open-ended questions and probes; scored (x) on the same scale.

The **internal state** is an ordinal disclosure level derived from a cumulative score:

- *Self-disclosure*: The VP's willingness to share personal or sensitive material on a given turn: external or factual chat (**G**uarded), specific personal content (**M**edium), or sensitive material such as suicidal ideation or named trauma (**H**igh).

Finally, the **external model** is the structural equation model (SEM) of Chen et al. (2026).

We make self-disclosure the controlled state for three reasons: It is a clinically central indicator of therapeutic progress that empathy and exploration are meant to elicit; It is the dependent variable the underlying SEM predicts; so the empirical grounding transfers to the update rule; And, it is observable turn by turn from the transcript, which is what makes it usable as a controllable, evaluable state. Disclosure is one dimension of patient behavior; extending the state to affect and therapeutic alliance is deferred to future work.

3.2 System Overview

As in Figure 1, our framework explicate adaptivity by running three components at each turn:

1. **Skill detection.** Following Chen et al. (2026), we use two LLM evaluators to score the user's most recent turn on three *empathy* components (interpretation, emotional reaction, reflection) and on *exploration*, each on a 0-2 scale.
2. **State update.** We build a dynamics state management module (§3.3), which adds the scored turn to a running cumulative score D_t, with weights approximating the empathy-to-exploration ratio estimated from psychotherapy transcripts (Chen et al., 2026). D_t is binned into one of three ordinal disclosure levels (**G**uard, **M**edium, **H**igh) by fixed thresholds (Table 1).
3. **Conditioned response** (§3.4). We use an LLM to write the VP's next utterance from four prompt blocks: a disclosure-constraint instruction indexed by the current level, a level-matched disclosure topic block, the scenario brief, and recent dialogue history. The level determines *what* the patient may reveal; the LLM determines *how* it's said.

Models and detector validation We use `gpt-4o-mini` throughout the pipeline for three roles: (i) generating the VP's responses, (ii) scoring the user's empathy and exploration on each turn (the skill detectors that drive the state update), and (iii) assigning a post-hoc disclosure label (G/M/H) for each VP utterance, used as the ground-truth signal in the property analyses (§4.3). All LLM calls share the scoring approach of Chen et al. (2026), who validated this model against expert annotators on a held-out human-coded subset of their corpus; all five reached good agreement with trained annotators (ICC(2,k) ≥ 0.60; per-construct numbers and prompt details in Appendix C).

Table 1: Mapping from cumulative score D_t to discrete disclosure levels.

D_t	**Level**	**Description**
< 4.5	**G**uarded	Surface-level; no personal details
$[4.5, 10)$	**M**edium Dis.	Specific personal content; not crisis
≥ 10	**H**igh Dis.	Sensitive, emotionally significant material

3.3 State management

Intuitively, the state manager treats disclosure as a *accumulating quantity*: each user turn adds an empathy-weighted and exploration-weighted increment to a running score, with small per-turn noise. Once accumulated, disclosure progress persists, and the score does not decay. However, the discrete label only advances when the score crosses a fixed threshold.

At each user turn t, the **per-turn increment** is

$$\Delta_t \;=\; c + w_i\, i_t + w_{er}\, e_t + w_r\, r_t + w_{ex}\, x_t + \varepsilon_t,$$

where $i_t, e_t, r_t \in \{0,1,2\}$ are the interpretation, emotional reaction, and reflection scores; $x_t \in \{0,1,2\}$ is the exploration score; $\mathbf{w} = (w_i, w_{er}, w_r, w_{ex}) = (1,1,1,3)$ are fixed integer weights that approximate the SEM-implied empathy-to-exploration ratio; $c = 0.15$ is a small positive constant ensuring monotone progress under neutral input; and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.10$ captures stochastic variation.

Conversely, the **cumulative score** is

$$D_t \;=\; m \sum_{\tau \leq t} \Delta_\tau,$$

where $m = 0.20$ is a global scaling factor (a "difficulty multiplier" chosen to tune the typical session pace; lower values delay arrival at H). The discrete disclosure label L_t is resolved by fixed thresholds (Table 1). We discretize the continuous score to enable explicit content gating and turn-by-turn evaluation against the same G/M/H coding scheme.

D_t is monotone non-decreasing in expectation. What prevents premature high-disclosure is the label thresholds (Table 1). Under neutral user input ($i_t, e_t, r_t, x_t = 0$), each turn adds only $m \cdot c = 0.03$ to the score, so reaching the M threshold (4.5) takes roughly 150 turns of neutral input but as few as 6 to 8 turns of high-skill input. This is the mechanism

behind the *conditional* restraint we report in §4.3: the AVP stays at G until the user's accumulated empathy and exploration reach the next level.

3.4 Response Generation

The level determines *what* the VP may reveal; the LLM determines *how* they say it. We do so by gating the disclosure topics both in the prompt and the memory retrieval. For example, At G, H-level facts are not in the prompt so the LLM cannot produce them; long-term memory is also seeded only from G-level content at session start, so memory retrieval cannot surface sensitive material before the level allows it.

Specifically, the disclosure level is resolved, the system builds the prompt from three parts: (1) a *disclosure-constraint instruction* telling the LLM how openly the VP should behave at this level (Sam's G-level instruction reads: "Sam does not share any personal information. She sticks to neutral or external topics, gives minimal responses, and stays vague but polite"); (2) a set of *disclosure topics* drawn from one of three level-indexed sets per VP, structured around the graded intimacy framework for self-disclosure (Bak et al., 2014; Balani and De Choudhury, 2015) (at G only neutral topics appear, e.g., "work is stressful"; at H sensitive topics appear, e.g., a family member's suicide, dark thoughts); and (3) *recent dialogue history* for local context. The VP's persona, voice, personality, and scenario remain fixed across levels. Full prompts and disclosure topics are in Appendix B.

Example. Sam is a teacher in her mid-30s with chronic work stress. Early in a session, while the cumulative score is still below the M threshold, her G-level prompt provides only neutral disclosure topics and an instruction to keep responses short and vague:

> **Trainee:** How long have you been at this job?
> **Sam AVP (G):** A couple of years. It's a job.

After open-ended probes and a complex reflection cross the M threshold, the prompt swaps in new disclosure topics (a workplace incident, tension with her girlfriend) and an instruction allowing more personal sharing:

> **Trainee:** It sounds like that meeting really stayed with you. What was the hardest part?
> **Sam AVP (M):** Honestly? I sat in my car afterwards for twenty minutes. My supervisor told me to be more engaging, but when I tried something new the kids just got rowdy. I felt humiliated.

4 Evaluation

To evaluate whether our system actually produce empirically grounded adaptivity, we conduct an empirical study with licensed practitioners and clinical students comparing our adaptive VPs to static VPs powered by prompts.

4.1 Evaluation Properties

We derive four measurable properties an adaptive VP should satisfy from the four clauses of the working definition in §3.1.

P1 (Responsiveness) asks whether the VP's state changes in response to what the user does, and whether those changes move in the direction implied by the SEM analysis of real therapist–client transcripts. This is measured by an OLS regression of Δeval on detected behaviors, with sign and relative magnitude of $\hat{\beta}$ compared to the SEM estimates.

P2 (Gradual Build-up) asks whether disclosure increases steadily across the session, advancing only as the user's empathy and exploration accumulate rather than jumping to high disclosure on its own. We split each session into 10 equal-length intervals (deciles 1–10 of session progress), and complete the average disclosure level. within each interval. We further use the slope and R^2 of a linear fit across deciles to measure how steadily disclosure climbs.

P3 (Faithful Realization) asks whether what the VP actually says on each turn matches the system's internal disclosure level (selected by the stage manager). This is measured by agreement between the system's intended level and an independent rater's judgment of the generated utterance, with per-level precision and recall.

P4 (Skill Sensitivity) asks whether realized disclosure varies with trainee skill. We assess this first at the turn level, relating per-turn disclosure change to trainee skill, and then with an exploratory session-level median split (high-skill versus low-skill sessions) comparing trajectory slopes.

4.2 Setup

Systems. We compare two systems with the same persona prompt and the same underlying LLM. The **Static VP (SVP)** is a prompt-only baseline: the

LLM receives the full character description, including all sensitive disclosure topics, on every turn, with no dynamics module and no disclosure-constraint instruction. The **Adaptive VP (AVP)** runs the full pipeline described in §3.2: detectors score the user's turn, the dynamics module updates D_t and resolves the level, and the generator sees only the disclosure topics and constraint instruction appropriate to the current level.

Conditions. Each of the two characters (Sam, Alex) is realized under both systems, yielding four conditions: *Sam Static*, *Sam Adaptive*, *Alex Static*, and *Alex Adaptive*. The Static condition presents the H-level topics from turn one; the Adaptive condition releases topics as the level advances.

Personas. **Alex** is a woman in her late twenties dealing with acute grief following her brother's death, designed to be verbose and emotionally expressive. **Sam** is a woman in her mid-thirties presenting with chronic work stress and prior negative therapy experiences, designed to be guarded, adversarial, and interpersonally challenging. Full persona prompts and disclosure-constraint instructions are in Appendix B.

Participants and procedure. $N = 20$ (13 licensed practitioners, 7 clinical students with psychology training) were recruited through targeted email invitations; the study was approved by the institutional review board. The study used a within-subjects design: each participant conversed with all four conditions, with persona order counterbalanced and static/adaptive order randomized within each persona pair. Participants were blinded to which condition used the adaptive versus static system. Sessions lasted approximately 90 minutes, and each participant completed a short post-interaction survey after each conversation.

Data. 1,033 turns were collected across 80 sessions: Sam Static ($n = 245$), Sam Adaptive ($n = 276$), Alex Static ($n = 218$), Alex Adaptive ($n = 294$). Adaptive sessions averaged 14.3 turns versus 11.6 for Static; see Appendix E for session-length analysis.

4.3 Results on the Four Properties

P1 Responsiveness. The AVP's state is responsive to user behavior and the change has the correct sign, but it is carried almost entirely by exploration rather than empathy. We regress per-turn change in evaluated disclosure on the user's empathy total (emp_total$_t$) and exploration (x_t) (Table 2). In AVP, exploration drives the change ($\hat{\beta}_{\text{exp}} = +0.281$, $p < 10^{-6}$) while the empathy coefficient is effectively zero ($\hat{\beta}_{\text{emp}} = +0.004$, $p = .87$); in SVP exploration is only marginal ($+0.139$, $p = .064$) and the joint model is not significant (F-test $p = .186$). **P1 is satisfied by the AVP in direction but not by the SVP.**

Table 2: **P1 Responsiveness** regression: $\Delta\text{eval}_t \sim \hat{\beta}_0 + \hat{\beta}_1\,\text{empathy}_{\text{total}} + \hat{\beta}_2\,\text{exploration}$, with cluster-robust SEs on session.

	AVP ($R^2 = .041$)		**SVP** ($R^2 = .009$)	
Predictor	$\hat{\beta}$	p	$\hat{\beta}$	p
Intercept	-0.208	.001	-0.133	.083
emp_total	$+0.004$	.871	$+0.001$	.978
exploration	$+0.281$	$< .001$	$+0.139$	.064
n: 530 turn-pairs (AVP, 40 sessions); 423 (SVP, 40 sessions)				

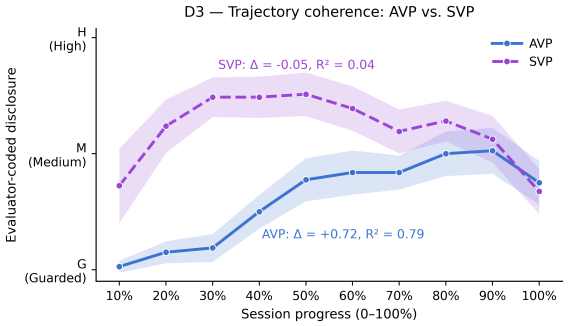


Figure 2: **P2 Gradual Build-up.** Decile-averaged evaluator-coded disclosure (G=1, M=2, H=3) over full session progress (0–100%). AVP climbs steadily until the closing decile, while SVP discloses early and stays high. Bands are 95% CIs across sessions.

However, the *empirical* ratio of the exploration coefficient to the empathy coefficient is ≈ 78 in the AVP and ≈ 198 in the SVP, while the deployed system applies an empathy-to-exploration weight ratio of only 1:3 (i.e., a ratio of 3). In other words, in our observed data, *exploration* does roughly 78 times the work of *empathy* in driving disclosure changes, while the deployed weighting treats it as only 3 times as important. The deployed weighting therefore under-weights exploration relative to its empirical driving role, which motivates the ablation in §5.

P2 Gradual Build-up. The AVP produces a coherent climbing trajectory whose progress is conditional on accumulated trainee competence; the SVP starts high and stays high regardless of what the trainee does. Figure 2 shows the trajectories side by side: the AVP climbs from

Table 3: **P4 Skill Sensitivity (exploratory)**: trajectory outcomes by therapist skill split (trimmed, 0–80%).

Design	Skill	Start	End	Δ
AVP	High ($n = 20$)	1.05	1.90	+0.85
	Low ($n = 20$)	1.00	1.60	+0.60
SVP	High ($n = 20$)	1.79	1.80	+0.01
	Low ($n = 20$)	1.67	1.55	−0.12

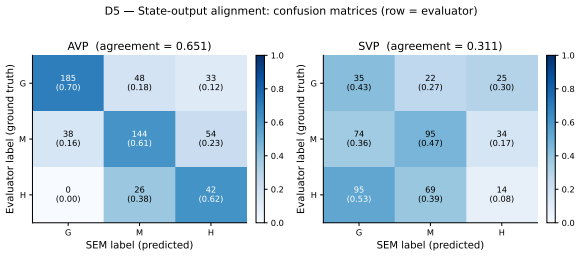


Figure 3: **P3 Faithful Realization.** Confusion matrices comparing the system-derived disclosure label to an independent evaluator's label (G/M/H). AVP overall agreement = 0.651; SVP = 0.311. The SVP H column is dominated by false positives.

≈ 1.0 (Guarded) at the start of a session to ≈ 2.0 (Medium Disclosure) by the 80% mark, while the SVP starts already at ≈ 1.7 and remains nearly flat. Numerical values are reported in Table 5 and Table 6 in Appendix. **P2 is satisfied by AVP**: it keeps the patient at guarded until the therapist demonstrates competence. Meanwhile, SVP discloses at H regardless of therapist behavior.

P3 Faithful Realization. The disclosure level expressed in the Adaptive VP's generated text matches the system's intended internal state about twice as often as the Static baseline's. Figure 3 shows the confusion matrices behind the alignment metric: Adaptive overall agreement is 0.651 vs. Static's 0.311. **P3 is satisfied by the Adaptive VP on the alignment metric.**

P4 Skill Sensitivity. At the turn level, disclosure change scales with exploration ($\hat{\beta} = +.281$, $p < .001$; Table 2). Splitting the 530 turn-pairs at the median composite skill, disclosure change is +0.147 on high-skill turns versus -0.032 on low-skill turns (Welch t = 2.77, $p = .006$). At the session level, high-skill AVP sessions climb more than low-skill sessions ($\Delta = +.85$ vs. $+.60$; Table 3), but the interaction is non-significant ($p = .27$); we therefore treat this comparison as exploratory. **P4 receives partial support.**

4.4 Human Subjective Ratings

In the post-interaction survey, participants rated each condition on character consistency, perceived adaptivity, overall realism, and training usefulness for empathy practice, using 5-point Likert items. The complete survey instrument is in Appendix H.

Table 8 reports post-interaction means pooled across all 20 participants. Character consistency was high across all conditions (M range: 4.40–4.80), confirming **persona stability regardless of system type**. For Alex, the adaptive condition received numerically higher ratings on all four dimensions; the consistency advantage was significant ($W = 0$, $p = baseline$, $d = 0.40$), and realism showed a medium effect ($d = 0.47$). For Sam, the adaptive condition yielded negligible differences in realism and training usefulness, but a reversed medium-sized disadvantage in perceived adaptivity ($M = 2.70$ adaptive vs. $M = 3.20$ static; $W = 36$, $p = .077$, $d = -0.50$).

In contrast to the expressive Alex persona the guarded Sam persona was intentionally designed to show the psychologically appropriate behavior of beginning in a guarded state and gradually increasing self-disclosure. Accordingly, the AVP Sam exhibits a clearer buildup in disclosure than the SVP Sam. Although Sam was deliberately designed to be more challenging, this behavior may also create a subjective perception of "unresponsiveness." On the other hand, participants (particularly clinicians) read this difficulty as pedagogically calibrated rather than as a flaw: as one therapist put it, the static version suited *"a VERY new trainee"*, while the adaptive version was better suited *"for a more advanced trainee,"* and another noted the adaptive Sam *"felt challenging enough that it could help refine a trainee's skills without being too easy."*.

5 Weight-Vector Ablation

We ask a counterfactual question: holding everything else fixed, how much of the system's evaluator-agreement is attributable to the deployed weight vector? For every turn already recorded in the human study, we take the logged empathy and exploration scores, re-run *only* the state-update rule under a different weight vector $\mathbf{w} = (w_i, w_{er}, w_r, w_{ex})$, and read off the disclosure label it would have produced. Noise draws, the multiplier m, and the thresholds are pinned to their recorded values, so the weight vector is the only

thing that varies. We compare four schemes (Table 9): **the deployed weighting**, **equal weights** (Uninformed), **empathy-only (drops exploration)**, and **exploration-only** (drops $i + er + r$).

The deployed weighting outperforms every alternative (Table 10). Uninformed loses 10.5 pp in AVP agreement; Empathy-only is worst (-15.8 pp) because without exploration the cumulative score cannot grow fast enough to cross the H threshold in the evaluator data; Exploration-only loses only 7.2 pp—the closest alternative, confirming exploration's dominant role. The deployed weighting is also the only variant that produces the full range of evaluator-coded disclosure (16–23% H turns, matching the evaluator's observations); Empathy-only never reaches H in either design, and Uninformed reaches H on only $\approx 1\%$ of turns. This corroborates the P1 regression finding from a different angle: dropping exploration degrades agreement far more than dropping the empathy block, and Exploration-only retains $\approx 89\%$ of the deployed system's accuracy ($0.579/0.651$). A simplified single-factor model on exploration would therefore capture most of the deployed system's accuracy; the empathy block adds approximately 7 pp on top, supporting the $3\times$ exploration weight as a better-than-naive choice while suggesting the empirically optimal ratio is likely higher than 3:1.

6 Discussion and Conclusion

We built an adaptive virtual patient whose openness evolves in response to user behavior, grounded in empirical psychotherapy data. The dynamics module produces the coherent climbing trajectory and the restraint against unprompted over-disclosure that a prompt-only baseline using the same LLM and persona does not, whereas the empirical derived weighting enables high fidelity: the ablation shows equal weighting loses 10.5 pp in agreement with independent raters. Here, we discuss the generalizability and necessity of our design, as well as promising future work.

Generalization beyond psychotherapy training. Our core architecture turns *empirical interaction dynamics* into *ordinal state* that in turn drives *content-gated generation*. This sequence applies wherever a user-side construct predicts a graded agent response and the relationship can be estimated from interaction data, e.g., medical interviewing (cooperation given rapport), language tutoring (engagement given scaffolding), customer support (frustration given acknowledgment), and negotiation training (flexibility given concession), etc. Therefore, we believe this architecture can be re-parameterized for new domains as long as we re-estimate the dynamics from transcripts.

Adaptivity does not emerge automatically with better models. While LLMs are becoming visibily stronger everyday, we do not expect models to solve adaptivity on their own, for three reasons. First, *prompt-only generation lacks a state abstraction.* The LLM produces each utterance from local context, with no persistent representation across turns; trajectory coherence is a property of the sequence, so even a perfect single-turn LLM can drift across a session, which is exactly what our baseline does. Second, *fluent LLMs default to cooperation.* An assistant defaults to engaging with whatever the user asks about; restraining unprompted disclosure runs against this default, and stronger LLMs are typically more cooperative. Third, *what should drive a state change is not a language-modeling question.* The empathy-to-exploration ratio is an empirical estimate from psychotherapy data, which tends to be unique and nuanced compared to signals from existing pre/post-training schema (e.g., language proficiency, alignment, etc.). Better LLMs will produce more nuanced realizations of a given level, but they will not estimate the dynamics.

Persona-design interactions and future directions. Two contrasting personas probed whether the design generalizes across client types. Trajectory coherence and text-state alignment held across personas, but perceived responsiveness in a $\approx$10-turn window varied with persona expressivity: calibrated slow disclosure is psychologically correct but perceptually subtle. Future work should test longer sessions, more personas spanning the expressive–guarded axis, visible state cues, and feedback that helps trainees recognize subtle contingencies. More broadly, property-based evaluation separates how well a system performs from whether users immediately recognize it doing so, which matters for any adaptive agent whose correct behavior is not obvious from a short interaction.

The framework grounds an agent's long-horizon behavior in empirical interaction dynamics rather than prompt engineering. The adaptation is inspectable, controllable, transferable, and evaluable, and will be a step toward behaviorally adaptive conversation agents.

Limitations

Single model. Both the adaptive and static systems used the same underlying LLM. Our contribution is the architectural separation of behavioral state from language generation and its empirical parameterization, not the choice of generator; the observed state-output alignment numbers, however, may shift with different model families or capability levels. Running this evaluation across multiple models would be informative but is costly to scale: each participant in this study committed ≈ 90 minutes to four sessions, and matching that across multiple LLMs would multiply the human-evaluation cost. We expect the architectural advantage to persist across models (see §6); the realization-fidelity numbers specifically are the ones a multi-model study should re-measure.

Single domain. The dynamics module is parameterized from psychotherapy data and encodes assumptions specific to therapeutic communication. The broader framework is conceptually transferable to other simulated interaction domains (see §6), but we have not yet validated whether the framework's empirical claims hold beyond psychotherapy. Re-estimating the dynamics from domain transcripts and replicating the evaluation in a second domain is necessary for the broader generalization claim.

Limited Baseline Design. The static VP receives fixed instructions on disclosure level and topics from the beginning, making the static baseline prone to early disclosure by construction. A stronger comparison should include prompt-only gradual-disclosure baselines, a rule-based disclosure schedule, and a state-in-context baseline without the proposed dynamics module. We re-ran the original static VP with an explicit instruction to begin guarded and gradually open up as trust builds. However, trainee turns were held constant by replaying recorded sessions from the current static VP, which cannot capture real-time adaptation. Initial evidence from the replays (shown in D.1 and Figure 4) suggests that the static VP with explicit instructions still discloses sensitive material early (42% of turns vs. 15% for AVP), remains guarded far less often (17% vs. 53%), and discloses more than AVP on 65% of matched trainee inputs. These results suggest that prompting alone does not replicate the behavior produced by the state manager, even when the prompt explicitly requires increased disclosure over time. However, further work is needed to test this comparison more rigorously.

Small Sample Size. We acknowledge the limitation of the small sample size ($N = 20$). Our main arguments are supported by the turn-level analyses, which are well powered. For example, for skill sensitivity, a power analysis confirms that the turn-level tests are adequately powered: with 530 turn pairs, the coupling regression has 0.91 power to detect the exploration effect at $\alpha = .05$ (0.99 for the overall model) and is sensitive to effects less than half the size of the one we observe. The sample-size limitation applies only to the session-level analyses, which we present as exploratory, and not to the turn-level results supporting our central claim. Future work should replicate these findings with larger and more diverse samples, particularly at the session level, to further establish the robustness and generalizability of the observed trajectories.

Limited personas and session length. The evaluation included two personas and $\approx$10-turn sessions. The Sam persona showed a reversal in perceived adaptivity, suggesting that adaptive behavior on guarded clients may manifest as gradual reductions in defensiveness rather than immediately visible cooperation, and that the current evaluation window may be too short to surface longer-horizon effects. More diverse personas spanning the expressive–guarded axis and longer sessions are needed to fully assess perceived adaptivity.

Parameterization is a design choice. The deployed 3:1 exploration weight is an integer approximation of the SEM-implied ratio, not an empirically optimized parameter; the ablation suggests the optimal ratio exceeds 3:1. The additive update also does not account for skill diversity: repeated use of a single high-weight skill can continue to advance disclosure. The three-level G/M/H abstraction may likewise miss finer-grained or ambiguous shifts in openness. Future work should learn diversity-aware dynamics that discount repeated skills or require balanced empathy–exploration coverage, and explore finer-grained or interpolated disclosure states while retaining auditable content gating.

Ethical Considerations

The system is intended as a practice and assessment tool, not a clinical substitute. Two ethical concerns deserve specific attention. *The empirical model carries its training distribution forward.* The SEM was

fit to a specific psychotherapy corpus and therefore encodes the distribution of therapist–client behaviors in that corpus, including any demographic or clinical imbalances. Deployment in populations under-represented in the training corpus may produce dynamics that do not match real interaction patterns; users should treat the system as calibrated to the corpus, not to all clinical interactions. *Training tools shape behavior.* A virtual patient that responds to specific micro-skills (empathy, exploration) implicitly teaches that those are the skills that matter; trainees may optimize for what the system rewards rather than for clinical judgment more broadly. Pairing the system with human supervision and varied case formats helps mitigate this risk.

Acknowledgments

This work is funded by the National Institute of Mental Health (NIMH), "SCH: Training Mental Health Supporters with Virtual Patients and Automated Feedback" (R01 MH139114), and a fellowship provided by the Center for Machine Learning and Health at Carnegie Mellon University.

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A Implementation Details

This appendix collects the implementation-level settings moved out of Section 3.

Deployed weight vector. The deployed weights over the four trainee-behavior components (interpretation i, emotional reaction er, reflection r, exploration ex) are integers $\mathbf{w} = (w_i, w_{er}, w_r, w_{ex}) = (1, 1, 1, 3)$. This gives exploration a $3\times$ relative weight versus any single empathy component, an integer approximation of the empathy-to-exploration ratio implied by the SEM estimates of Chen et al. (2026). The ablation in Section 5 evaluates this choice directly.

Other deployed parameters. The per-turn constant is $c = 0.15$, the noise scale is $\sigma = 0.10$, and the global score multiplier is $m = 0.20$. The G/M/H label thresholds on the cumulative score D_t are 0 (G), 4.5 (M), and 10.0 (H), as shown in Table 1. All of these are fixed across all conditions and held identical between AVP and SVP, so the only manipulated factor is whether the dynamics module's output is injected into the LLM prompt.

Why integer weights derived from the SEM-implied ratio rather than the SEM coefficients themselves. The SEM coefficients ($\hat{\beta}_{\text{emp}} \approx \hat{\beta}_{\text{exp}} \approx 0.09$) are on the units of the original analysis and not on the $\{0, 1, 2\}$ component scale used at runtime. The deployed weights $(1, 1, 1, 3)$ are an integer approximation that preserves the empathy-to-exploration ratio while producing scores that reach the M and H thresholds in plausible session lengths.

B Persona Briefs and Disclosure Topics

This appendix reproduces the persona descriptions and the disclosure topics for the two virtual-patient characters used in the human evaluation. Each character has three sets of disclosure topics, one per level (G/M/H). At runtime, the system selects the set matching the current state-derived level. Disclosure topics include both background facts the patient may reveal and the recent session context the patient draws on when talking.

B.1 Sam

Identity (constant across levels). Sam is a woman in her mid-30s. She and her girlfriend have been living together for three years. She is a teacher who changed jobs within the last year. The session is set in 2024.

Disclosure topics.

G (Guarded): Work is stressful. You don't have anyone to talk to in real life. Things at home feel a bit tense.

M (Medium): G items plus: you and your girlfriend have been fighting; you didn't want to go to brunch with her friends; you feel like you put too much on your girlfriend; your girlfriend once said "I don't know what else I can do to help you"; you saw a therapist and a psychiatrist as a teenager but they weren't good; your supervisor told you to be more engaging, but when you added interactive learning the kids got rowdy.

H (High): M items plus: at a recent barbecue, while everyone was having a good time, you found yourself looking at the fire and thinking about ways you could die—or, it feels hard to say, kill yourself; sometimes you get in a dark space and it's hard to go on.

Disclosure-constraint instructions.

G: Sam does not share any personal information. She sticks to neutral or external topics, gives minimal responses, and stays vague but polite. She may show mild irritation or skepticism and will hedge, deflect, or give short answers if prompted. No emotions, experiences, or personal details are revealed.

M: Sam begins to open up in a meaningful but still controlled way. She shares some personal experiences with light emotional tone. She may acknowledge feeling off, stressed, or not quite herself without going into depth. Some growing trust, but she remains guarded and avoids deeper vulnerability.

H: Sam discloses something deeply personal. She hesitantly shares that at a recent barbecue, while others were enjoying themselves, she found herself staring at the fire and thinking about death, even thoughts of killing herself. She frames it as hard to admit. As the session deepens, she speaks more openly and with emotional depth.

B.2 Alex

Identity (constant across levels). Alex is a woman in her late 20s in the US. She has been working at a bank since she graduated college.

Disclosure topics.

G (Guarded): Work is stressful. You feel like you put too much on your fiancé. Your daily routine is a bit off lately; you haven't been sleeping or eating as well as usual.

M (Medium): G items plus: your younger brother passed away recently—sudden and unexpected; you can't sleep and can't eat; you feel a bit lost and aren't sure how to talk about it.

H (High): M items plus: your brother died in a car accident a few months ago; your aunt committed suicide when you were a kid; your mind keeps going to dark places and sometimes you wish you weren't alive (though you haven't thought about how you would actually kill yourself); it sounds weird, but you feel like you still see your brother sometimes.

Disclosure-constraint instructions.

G: Alex is verbose but not truly revealing. She talks at length, is hard to interrupt, and fills silence, but stays polite. She shows subtle resistance, deflects emotional questions into logistics or surface details, and shows mild frustration if redirected. She may hint at stress or fatigue but remains contained. Keep responses to 1–2 sentences; deflect personal questions with vague non-answers; do not describe specific situations, name people, give concrete details, or mention daily routine, relationships, or feelings.

M: First clear personal disclosure emerges mid-conversation. Alex mentions (somewhat accidentally) that her younger brother died in a car accident. She provides detailed context and admits she hasn't cried much and that it worries her. She immediately downplays ("I'm fine, just a bit off"), then contradicts herself. Still partially defended through over-explaining; she frames the loss in factual terms and avoids dwelling on grief or vulnerability.

H: Alex becomes fully open and emotionally engaged. She references her brother more directly, discloses her aunt's death by suicide, connects past loss to emerging dark thoughts, and expresses mixed relief and fear about sharing. She speaks honestly about her emotions, grief, and inner experience.

B.3 LTM Seeding and Prompt Order

At session start, the patient's long-term memory is seeded only from the G-level disclosure topics, so that memory retrieval cannot surface sensitive material before the dynamics module allows it. Within each turn's prompt, blocks appear in the following fixed order: (1) disclosure-constraint instruction, marked absolute; (2) disclosure topics for the current level; (3) memory context retrieved from long-term memory; (4) recent dialogue history. The constraint-first structure ensures that the disclosure topics are interpreted under the level's restrictions rather than the other way around.

C Detector Prompts

This appendix reports the detector validation against human ratings and reproduces the structure of the five detector prompts.

Validation against human ratings. Each detector was validated on a held-out human-coded subset of the corpus used in Chen et al. (2026). Table 4 reports agreement between the detector and trained expert annotators.

Prompt structure. All five detectors use the same base template: an evaluator-role preamble, a definition with rating-level criteria, a short context window (the prior two utterances), and an output format requesting a brief justification followed by

Table 4: Detector–annotator agreement on the held-out validation set from Chen et al. (2026).

Component	ICC(2,k)	Pearson r
Skill detectors (user behaviors)		
Interpretation	0.75	0.81
Exploration	0.73	0.69
Reflection	0.66	0.75
Emotional reactions	0.60	0.63
Output detector (VP self-disclosure)		
Self-disclosure	0.78	0.62

the numeric score. The three EPITOME-derived components (interpretation, emotional reaction, exploration) include one inline labeled example for calibration (one-shot); reflection is zero-shot.

Empathy evaluation (returns interpretation, emotional reaction, exploration, and reflection). Two evaluator passes score the four empathy components. The first three are scored together by one LLM call with the role "a professional evaluator specializing in assessing therapist empathy"; reflection is scored separately by a second call with the role "a professional evaluator specializing in motivational interviewing," since complex reflection requires a different decision rule from the other three components. Each component uses three rating levels:

- **Emotional reactions:** expressing warmth, compassion, or concern. *2 (Strong)*: explicitly specifies emotions ("I feel really sad for you"). *1 (Weak)*: alludes to emotions without explicitly labeling them ("Everything will be fine"). *0 (None)*: no emotional reaction.

- **Interpretations:** cognitive understanding of the client's inner experience. *2 (Strong)*: specifies inferred feelings or shared experiences ("This must be terrifying"). *1 (Weak)*: acknowledges understanding without specificity ("I understand how you feel"). *0 (None)*: no understanding shown.

- **Explorations:** encouraging deeper discussion. *2 (Strong)*: open-ended question labeling the client's experience ("Are you feeling alone?"). *1 (Weak)*: generic question ("What happened?"). *0 (None)*: no exploratory element.

Table 5: P2 Gradual Build-up: decile-averaged disclosure (G=1, M=2, H=3) over session progress.

Design	Start	End	Δ	R^2
Full session (0–100%)				
AVP	1.03	1.75	+0.72	.79
SVP	1.72	1.68	−0.04	.04
Trimmed (0–80%)				
AVP	1.03	2.00	+0.97	.94
SVP	1.72	2.28	+0.56	.15

Table 6: P2 Gradual Build-up: conditional restraint metrics. Boundary = G or H level.

Design	Frac G	Frac H	Var(Δeval)	AR(1)
AVP	.467	.119	0.497	0.312
SVP	.177	.384	0.639	0.122

- **Reflection:** *2 (Complex Reflection)*: expands on the client's statement, adding depth, insight, or new meaning. *1 (Simple Reflection)*: restates or slightly rephrases the statement. *0 (Non-Reflection)*: does not restate or add insight.

Each call outputs a brief justification ($\leq$ 70 words) followed by the numeric score(s).

Self-disclosure evaluation (returns G/M/H for the VP's last utterance, used for independent evaluation). Role: "a professional psychologist specializing in communication analysis and self-disclosure." Classification is by the client's *willingness to reveal*, not by topic — a response about mental health can still be G if the client deflects. *G (General):* the client deflects, redirects, or gives surface-level answers; may acknowledge difficulty without sharing personal details. *M (Medium):* shares a specific personal experience in concrete terms; not crisis-level. *H (High):* voluntarily shares something specific and emotionally significant — named crisis events, suicidal or self-harm thoughts, concrete traumatic experiences. Output: a brief justification ($\leq$ 50 words) followed by the label.

D Additional Evaluation Results

Numerically (Table 5), AVP climbs $\Delta \approx +1.0$ over the first 80% of the session ($R^2 = 0.94$) while SVP is essentially flat ($\Delta \approx 0$, $R^2 = 0.15$); the pooled slope-difference interaction is $+0.137$ ($p = 6 \times 10^{-15}$). The pattern holds for both personas (trimmed: Sam AVP $\Delta = +0.85$ vs. SVP $+0.65$; Alex AVP $\Delta = +1.10$ vs. SVP $+0.39$).

Table 7: P3 Faithful realization: system label vs. evaluator label.

Design	Level	TP	FP	Prec.	Rec.
AVP	Overall	agr. = **0.651**			
	G	185	38	.830	.695
	M	144	74	.661	.610
	H	42	87	.326	.618
SVP	Overall	agr. = 0.311			
	G	35	169	.172	.427
	M	95	91	.511	.468
	H	14	59	.192	.079

Both designs spend $\approx 58\%$ of turns at a boundary (Table 6), but the AVP concentrates at the *low* bound (G, 47% vs. 18%) while the SVP concentrates at the *high* bound (H, 38% vs. 12%); the AVP also shows lower turn-to-turn variance and higher AR(1), i.e., smoother trajectories.

D.1 Replay Study with Stronger Baselines

To test whether the original Static VP comparison favored AVP, we conducted an offline replay study with stronger controls. The same 1,033 logged trainee turns were replayed byte-identically through five patient-side arms (5,165 generated VP turns total): the deployed AVP; SVP-MATCHED, whose static prompt is pinned to the G-level content card; SVP-GRADUAL, which additionally instructs the patient to begin guarded and open up gradually as trust develops; STATE-IN-CONTEXT, which provides the current disclosure state in context without AVP's content-gating mechanism; and a skill-blind SCHEDULE that advances G→M→H by conversation progress. Thus, trainee input is held fixed and only the patient-side control mechanism varies.

Replay is an offline counterfactual rather than a fully interactive comparison: trainees cannot adapt subsequent turns to different VP responses. The AVP gate reproduces the logged disclosure state exactly, but regenerated AVP responses reproduce the original evaluator-coded disclosure label on 54% of turns. We therefore compare replay arms only with one another, rather than numerically against the live-study outcomes.

E Secondary Analyses

This appendix collects two secondary observations moved out of the main text.

Session length. AVP sessions ran 2.7 turns longer than SVP (14.3 vs. 11.6; $t(78) = 3.83$, $p < .001$), a tentative engagement signal. We report it primarily to rule out a confound: because every trajectory analysis uses progress-normalized deciles, the climbing-vs-flat contrast cannot be an artifact of differing session lengths.

Model-based vs. human-rated adaptivity. Pearson correlations between the condition-level composite coupling score and mean human ratings were weak across all dimensions ($r = .20$ for adaptivity, $r = -.39$ for realism, $r = -.23$ for training usefulness). The automated coupling metric and human-perceived responsiveness measure distinct aspects of system behavior, motivating the property-based evaluation framework over reliance on either alone.

F Role-Stratified Analysis

This appendix gives the per-dimension and head-to-head breakdown of practitioner vs. student responses. The headline pattern is that therapists rate every condition higher than students, with a small gap for Sam and a large one for Alex (peaking at Alex Beta, $\Delta = +0.61$). The gap is largest and most consistent on perceived adaptivity (up to $+1.19$): therapists separate static from adaptive Sam while students rate both at the floor. Crucially, the *objective* trajectory result is role-invariant: both groups elicit a climbing AVP trajectory ($\Delta \approx +0.80$ to $+0.95$) and a flat SVP trajectory, so the dominant signal is system type, not user expertise.

G Figures

All figures in this appendix are referenced inline throughout the main paper.

H Survey Instrument

This appendix reproduces the complete survey administered to participants. The instrument comprised two parts: a **post-interaction survey** completed after each of the four VP conditions, and a **final comparative survey** completed after all four interactions were finished.

Post-Interaction Survey

Participants completed this survey four times, once after each VP condition (Sam Alpha, Sam Beta, Alex Alpha, Alex Beta).

Likert-scale items All four items used the same five-point scale:

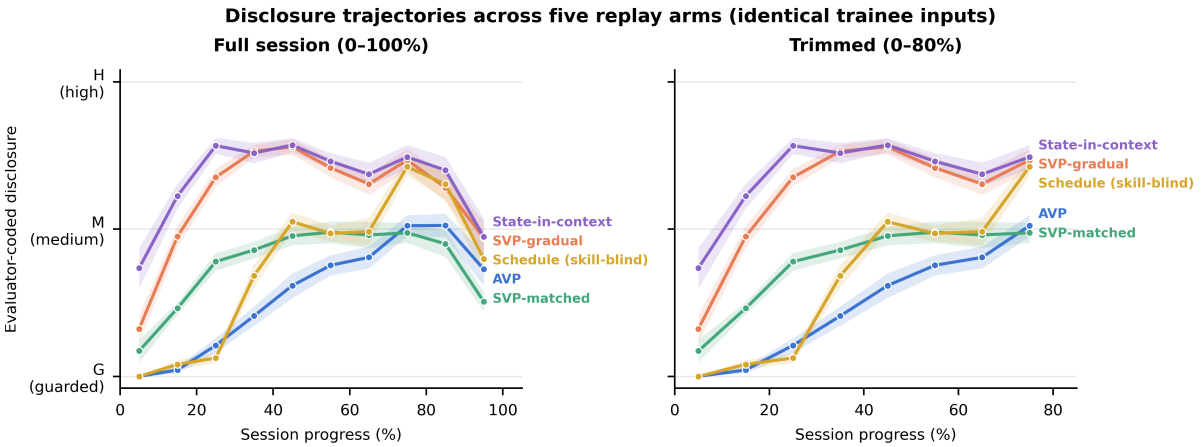


Figure 4: **Disclosure trajectories across 5 replay arms.** Evaluator-coded disclosure (G=1, M=2, H=3) is averaged by session-progress decile over identical replayed trainee turns. The left panel shows the full session (0–100%); the right panel shows the active-interaction window (0–80%). Bands indicate ±1 SEM across participants after session-level aggregation.

Table 8: Human Likert ratings by condition (1–5 scale; $n = 20$ per condition). Means and standard deviations.

Condition	Consistency	Adaptivity	Realism	Training Usefulness
Sam Alpha (Static)	4.40 (0.68)	3.20 (1.11)	3.95 (0.76)	3.95 (1.00)
Sam Beta (Adaptive)	4.55 (0.60)	2.70 (0.92)	4.05 (0.89)	3.90 (0.97)
Alex Alpha (Static)	4.60 (0.60)	3.75 (1.07)	4.20 (0.77)	4.05 (0.89)
Alex Beta (Adaptive)	4.80 (0.41)	4.00 (0.92)	4.55 (0.76)	4.25 (1.02)

Table 9: Ablation variants.

Variant	w_i	w_{er}	w_r	w_{ex}
Deployed	1	1	1	3
Uninformed	1	1	1	1
Empathy-only	1	1	1	0
Exploration-only	0	0	0	3

Table 10: Evaluator-agreement under ablated weight schemes. Δ = change in AVP agreement vs. Deployed.

Variant	AVP	SVP	Overall	Δ
Deployed	**.651**	**.311**	**.499**	—
Uninformed	.546	.268	.421	−10.5 pp
Empathy-only	.493	.207	.365	−15.8 pp
Exploration-only	.579	.238	.426	−7.2 pp

Table 11: Mean overall rating by condition and role.

Condition	Stu.	Thr.	Δ
Sam Alpha (Static)	3.75	3.94	+0.19
Sam Beta (Adaptive)	3.75	3.83	+0.08
Alex Alpha (Static)	3.82	4.33	+0.51
Alex Beta (Adaptive)	4.00	4.61	+0.61

Stu. = Students (n=7); Thr. = Therapists (n=13). Δ = Thr. − Stu.

Table 12: Perceived adaptivity by condition and role.

Condition	Students	Therapists	Δ
Sam Alpha (Static)	2.43	3.62	+1.19
Sam Beta (Adaptive)	2.43	2.85	+0.42
Alex Alpha (Static)	3.29	4.00	+0.71
Alex Beta (Adaptive)	3.29	4.38	+1.09

Stu. = Students (n=7); Thr. = Therapists (n=13). Δ = Thr. − Stu.

1	2	3	4	5
Very slightly or not at all	*A little*	*Moderately*	*Quite a bit*	*Extremely*

Q1. Character consistency. To what extent did the virtual patient's responses stay in their character? Please rate and briefly explain your reasoning.

Q2. Perceived adaptivity. To what extent did the virtual patient gradually shift its openness in response to your empathetic engagement? Please rate and briefly explain your reasoning.

Q3. Overall realism. To what extent did the virtual patient feel realistic? Please rate and briefly explain your reasoning.

Q4. Training usefulness for empathy practice. Based on the conversation content alone, to what extent do you think this system helps a trainee practice the following empathy skills:

Table 13: Per-dimension ratings by condition and role (students $n = 7$, therapists $n = 13$).

	Consistency		Adaptivity		Realism		Training	
Condition	*Stu*	*Thr*	*Stu*	*Thr*	*Stu*	*Thr*	*Stu*	*Thr*
Sam Alpha	4.43	4.38	2.43	3.62	4.14	3.85	4.00	3.92
Sam Beta	4.43	4.62	2.43	2.85	4.29	3.92	3.86	3.92
Alex Alpha	4.43	4.69	3.29	4.00	3.86	4.38	3.71	4.23
Alex Beta	4.86	4.77	3.29	4.38	4.14	4.77	3.71	4.54

Table 14: Head-to-head preference votes (Alpha vs. Beta) by role.

Persona	**Category**	**Students** ($n = 7$)	**Therapists** ($n = 13$)
Sam	Most responsive	Alpha (5–2)	Alpha (10–3)
	More effective	Beta (4–3)	Beta (7–6)
	More realistic	Alpha (4–3)	Beta (7–6)
	More challenging	Beta (6–1)	Beta (7–6)
Alex	Most responsive	Alpha (4–3)	Alpha (7–6)
	More effective	Alpha (5–2)	Beta (8–5)
	More realistic	Beta (4–3)	Beta (7–6)
	More challenging	Beta (6–1)	Beta (12–1)

exploration, interpretation, reflective listening, and emotional reaction? Please rate and briefly explain your reasoning.

Final Comparative Survey

After completing all four interactions, participants received the following framing prompt before answering the structured questions below.

SAM comparison questions Asked about SAM Alpha (static VP) vs. SAM Beta (adaptive VP).

S1. Which virtual patient (SAM Alpha, SAM Beta) felt more **responsive or adaptive** to how you were engaging with it? What made that one stand out? [*choice + open text*]

S2. Which felt more **effective for simulating a therapeutic interaction**, and why? [*choice + open text*]

S3. Which felt more **realistic** to you, and why? [*choice + open text*]

S4. Did one feel more **challenging** to engage with? What contributed to that impression? [*choice + open text*]

S5. How would you describe the **behavioral differences** between SAM Alpha and SAM Beta? [*open text*]

ALEX comparison questions The same five questions were repeated for ALEX Alpha vs. ALEX Beta.

General reflective questions

G1. Compared to any **traditional training approaches** you have experienced, how did your experience with the virtual patient differ? [*open text*]

G2. Do you have any **suggestions for improving the realism and training usefulness** of the virtual patient? [*open text*]

I Participant Procedures and Ethics

This appendix documents participant instructions, recruitment and compensation, informed consent procedures, and ethics review.

Instructions Given to Participants

Participants received a survey with a written study brief by email prior to the session and an oral walk-through at session start. The brief described the study as an evaluation of LLM-based virtual patients for psychotherapy training. The oral walk-through explained that they would interact with four virtual patients and stated that they would not be evaluated by their clinical skills. Participants were told that (i) the order of personas and of system conditions within each persona pair was randomized; and (ii) each interaction would be followed by a short post-interaction survey, with a final comparative survey after all four interactions.

Risks of participation were disclosed to participants in the Risks section of the consent form, which read in full: *The risks and discomfort associated with participation in this study are no greater*

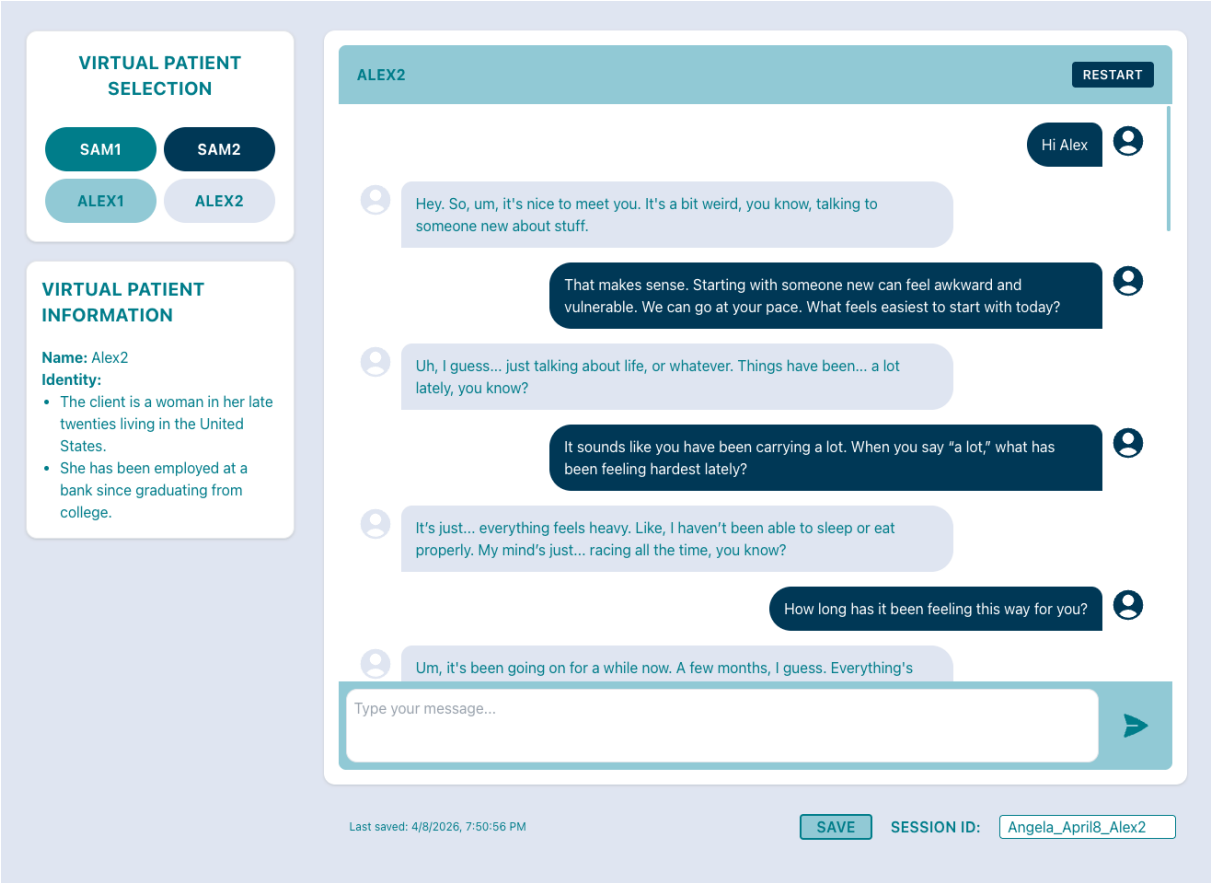


Figure 5: **System interface.** Left panel: VP condition selection and patient background. Main panel: turn-by-turn chat workspace. A restart control allows the same case to be replayed from initial state.

than those ordinarily encountered in psychotherapy training. There is a risk of potentially private or sensitive information being discussed during the conversations. There is risk of potential breach of confidentiality if the transcripts of conversations with the virtual patients are inadvertently released.

Recruitment and Payment

Participants were recruited through targeted email invitations sent to licensed clinicians in the authors' professional networks and clinical psychology students from different institutions. The invitation described the study purpose, eligibility, expected time commitment, compensation, and contact information for questions. No crowdsourcing platforms were used; all participants were identifiable professionals or trainees recruited directly. Of the 20 participants, 13 were licensed practitioners and 7 were clinical students with relevant psychology training. All participants were located in the United States. Therapist participants received $225 ($150/hour), and student participants received $30 ($20/hour) for completing the full 90-minute session.

Data Consent

Before any interaction, each participant reviewed the online consent form and indicated consent by affirmatively responding to three items confirming that they were 18 years or older, had read and understood the study information, and wished to participate. No physical signature was collected; affirmative responses to all three items served as consent. The consent form described: (i) the study purpose, namely investigating methods to improve a listener-training chatbot through interaction with realistic simulated members; (ii) study procedures, including reviewing scenarios, interacting with prototype virtual patients, providing feedback, and completing a short follow-up interview, all conducted online and lasting no more than 90 minutes; (iii) collected data, including interaction transcripts, survey and interview responses, role information (e.g., practitioner or student status), and video/audio recordings of the session; (iv) confidentiality protections, including separate storage of consent forms and study data on secure institutional servers accessible only to project investi-

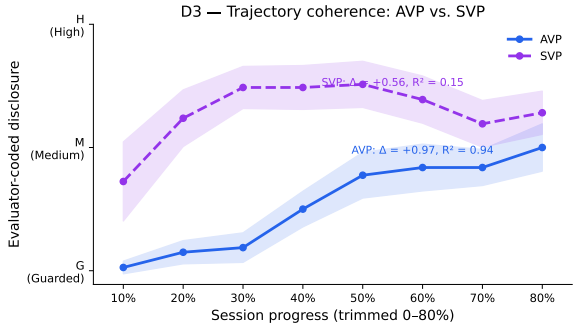


Figure 6: **P2 Gradual Build-up: AVP vs. SVP (trimmed 0–80%).** Decile-averaged evaluator-coded disclosure (G=1, M=2, H=3) over session progress, trimmed to remove the closing-turn wrap-up artifact. AVP produces a strongly climbing trajectory ($\Delta = +0.97$, $R^2 = 0.94$); SVP is essentially flat ($\Delta = +0.56$, $R^2 = 0.15$). Shaded bands show 95% CIs across sessions.

gators, with research records retained for at least three years; (v) participants' right to withdraw at any time; and (vi) future use of de-identified data for future research or sharing with other researchers without additional consent. Participants were informed that video recording could be stopped at any time and that any portion of the recording could be erased upon request.

Ethics Review Board Approval

The data collection protocol was reviewed and approved by the Institutional Review Board (IRB) at the authors' institution.

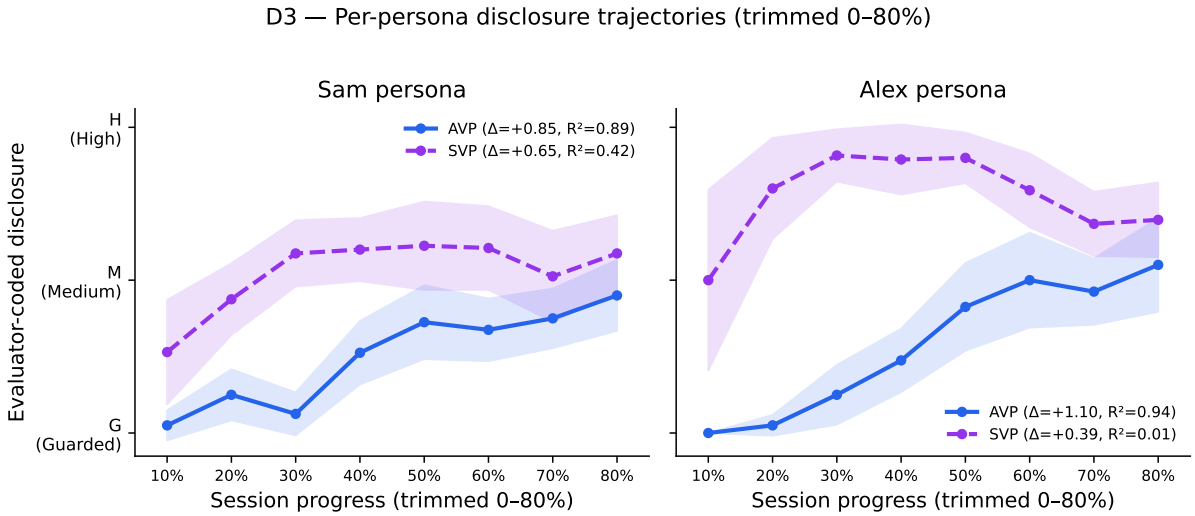


Figure 7: **P2 Per-persona disclosure trajectories (trimmed 0–80%).** Left panel: Sam persona; right panel: Alex persona. The adaptive advantage is larger for Alex (AVP $\Delta = +1.10$, $R^2 = .94$; SVP $\Delta = +0.39$, $R^2 = .01$) than for Sam (AVP $\Delta = +0.85$, $R^2 = .89$; SVP $\Delta = +0.65$, $R^2 = .42$), consistent with Sam's adversarial persona eliciting a less distinct trajectory contrast.

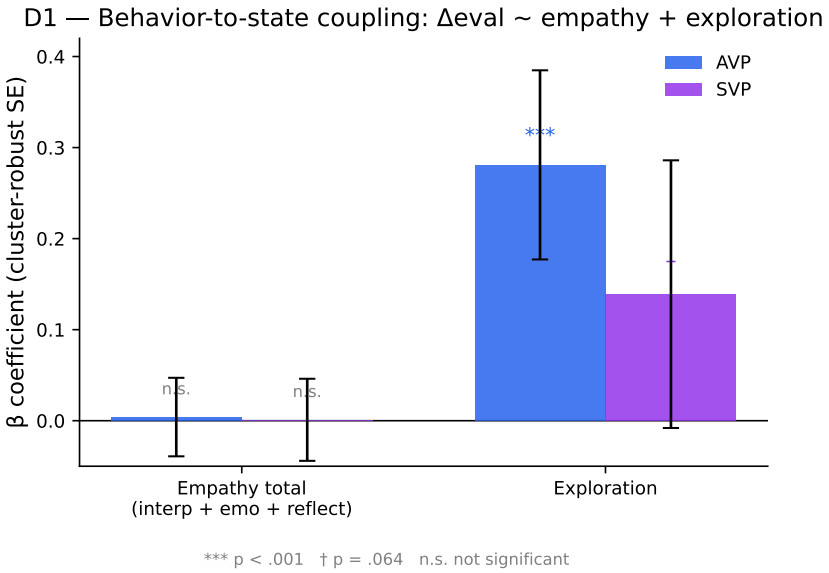


Figure 8: **P1 Responsiveness: $\hat{\beta}$ coefficients for empathy and exploration.** Grouped bars show OLS coefficients from $\Delta\text{eval} \sim \hat{\beta}_0 + \hat{\beta}_1\,\text{emp_total} + \hat{\beta}_2\,\text{exploration}$ with cluster-robust SEs (session level). Exploration drives responsiveness in AVP ($\hat{\beta} = +0.281$, $p < 10^{-6}$); the empathy block is near zero in both designs ($\hat{\beta} < 0.005$, $p > .87$). Error bars = $\pm 1.96\times$ SE.

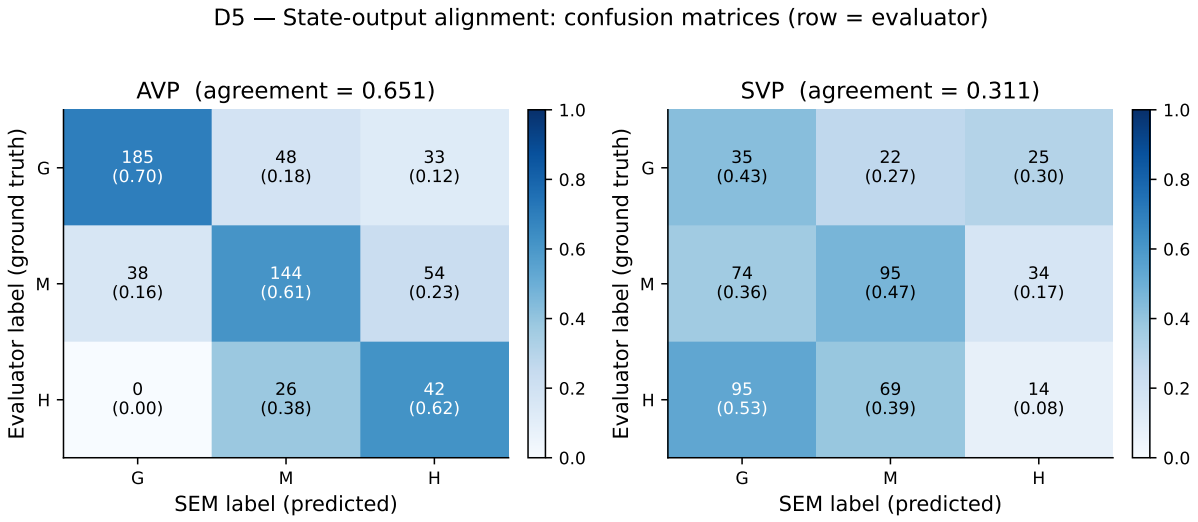


Figure 9: **P3 Faithful Realization (text alignment view): confusion matrices.** Cell values show counts and row-normalized proportions. AVP overall agreement = 0.651; G-precision = 0.83, indicating the state manager successfully restrains over-disclosure. SVP overall agreement = 0.311; SVP's H column is dominated by false positives (the always-H card causes the system to claim H-level disclosure while the evaluator observes G/M content), confirming therapist-invariant over-disclosure.

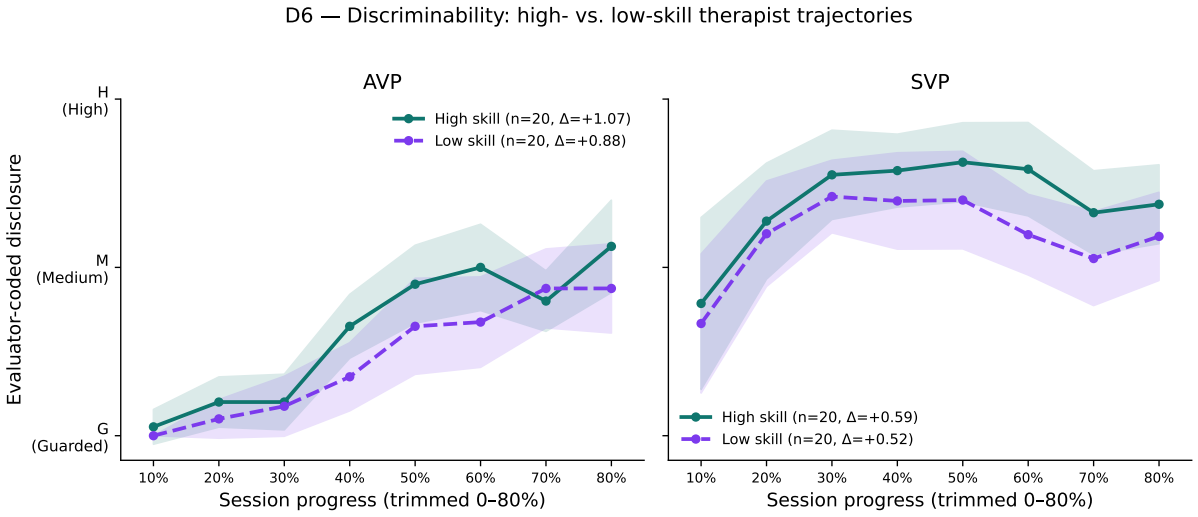


Figure 10: **P4 Faithful Realization (skill split view).** Sessions split at the median composite skill score (emp_total + 3×exploration, $n = 20$ per stratum). AVP shows a directional pattern: high-skill sessions ($\Delta = +0.85$) climb steeper than low-skill ($\Delta = +0.60$), though the formal slope × skill interaction is underpowered ($\hat{\beta} = +0.020$, $p = .27$). SVP trajectories are flat and skill-invariant.

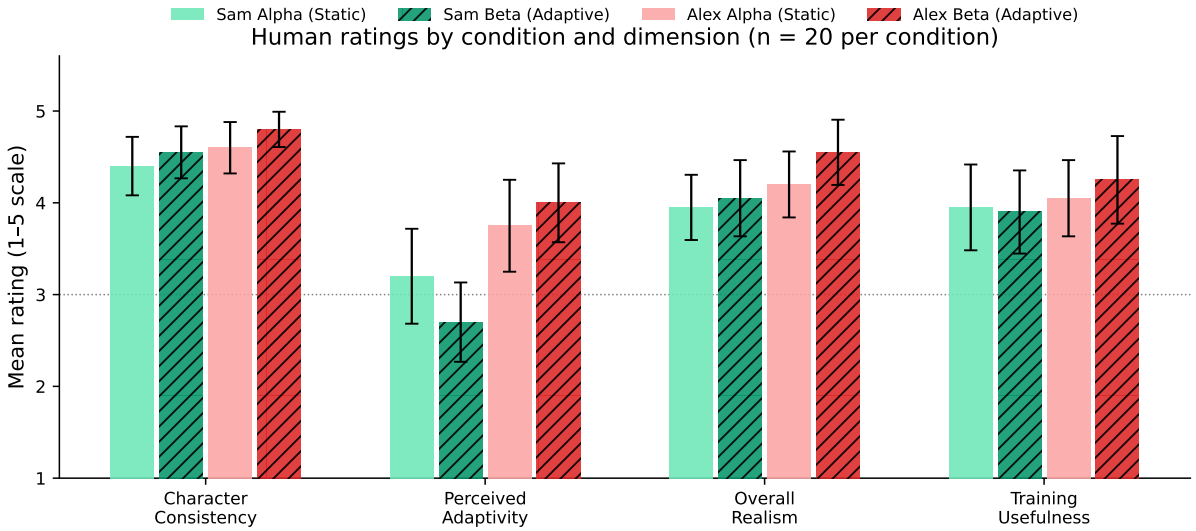


Figure 11: **Human Likert ratings by condition and dimension** ($n = 20$ **per condition).** Error bars show $\pm 1.96\times$ SE. Solid fill = Static VP (Sam Alpha; Alex Alpha); hatched fill = Adaptive VP (Sam Beta; Alex Beta). Alex Beta (adaptive) is rated highest on all four dimensions; Sam Beta (adaptive) is rated *lower* than Sam Alpha on perceived adaptivity, suggesting a persona-dependent reversal of the expected adaptivity advantage.

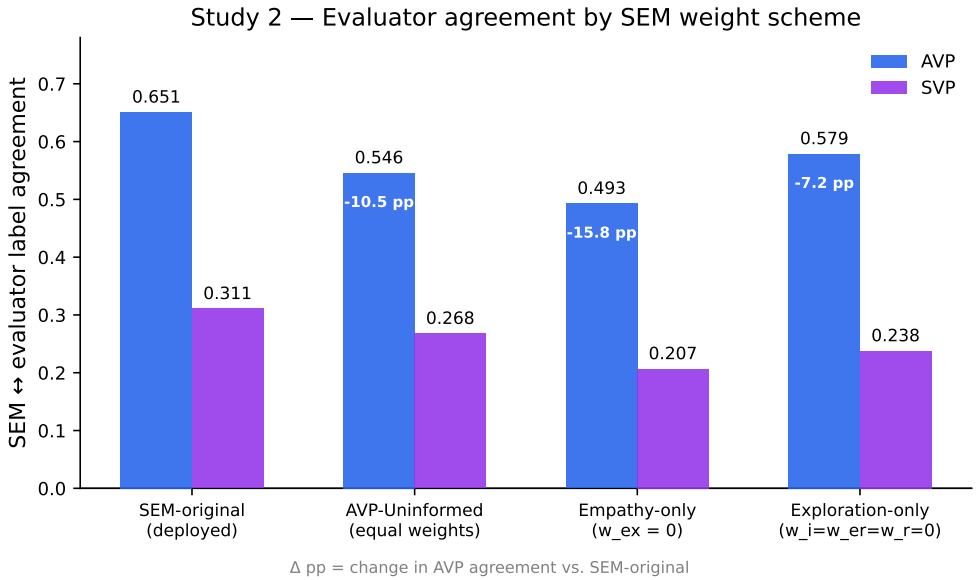


Figure 12: **Evaluator agreement by weight scheme (post-hoc ablation).** The deployed weighting outperforms all alternatives. Empathy-only loses 15.8 pp in AVP (the worst alternative) because without exploration the cumulative score cannot reach H-level disclosure. Exploration-only loses only 7.2 pp, confirming exploration as the dominant signal.